

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 3: Pair of Linear Equations in Two Variables

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Elimination method

1. The pair of equations $2x - 3y + a = 0$ and $2x + 3y - b = 0$ has the solution $x = 1, y = 2$. Find a and b , and state how they are related. **CBSE 2025, 1 mark**

SOLUTION

First equation: $2(1) - 3(2) + a = 0$, so $a = 4$.

Second equation: $2(1) + 3(2) - b = 0$, so $b = 8$.

Answer: $a = 4, b = 8$, so $b = 2a$

2. Solve the pair $7x - 2y = 5$ and $8x + 7y = 15$, and verify the solution. **CBSE 2024, 2 marks**

SOLUTION

Multiply the first equation by 7 and the second by 2: $49x - 14y = 35$ and $16x + 14y = 30$

Add: $65x = 65$, so $x = 1$. From $7x - 2y = 5$: $7 - 2y = 5$, so $y = 1$.

Verification: $7(1) - 2(1) = 5$ and $8(1) + 7(1) = 15$. Both are true.

Answer: $x = 1, y = 1$

2 Word problems

3. A person puts money into two schemes, A and B, which pay simple interest at 8% and 9% a year. The interest for one year comes to ₹1,860. If the two amounts had been swapped between the schemes, the interest would have been ₹20 more. How much was put into each scheme?

CBSE 2025, 5 marks

SOLUTION

Let ₹ x be invested in scheme A and ₹ y in scheme B.

Interest as invested: $\frac{8x}{100} + \frac{9y}{100} = 1860$, so $8x + 9y = 186000 \dots (1)$

Interest after swapping: $\frac{9x}{100} + \frac{8y}{100} = 1880$, so $9x + 8y = 188000 \dots (2)$

Adding (1) and (2): $17(x + y) = 374000$, so $x + y = 22000 \dots (3)$

Subtracting (1) from (2): $x - y = 2000 \dots (4)$

Adding (3) and (4): $2x = 24000$, so $x = 12000$ and $y = 10000$.

Check: 8% of 12000 + 9% of 10000 = 960 + 900 = 1860.

Answer: ₹12,000 in scheme A and ₹10,000 in scheme B

4. Three years ago Rashmi's age was three times Nazma's age. Ten years from now Rashmi's age will be twice Nazma's. Find their present ages. CBSE 2024, 3 marks

SOLUTION

Let Rashmi be x years old and Nazma y years old now.

Three years ago: $x - 3 = 3(y - 3)$, so $x = 3y - 6$.

Ten years later: $x + 10 = 2(y + 10)$, so $x = 2y + 10$.

$3y - 6 = 2y + 10$ gives $y = 16$, and $x = 42$.

Answer: Rashmi is 42 years old and Nazma is 16 years old.

3 Graphical method and consistency

5. A graph shows the lines of two linear equations. The lines are not parallel, but they do not meet in the part of the plane that is drawn. The pair of equations is: (a) consistent, with a unique solution (b) consistent, with infinitely many solutions (c) inconsistent (d) inconsistent, but made consistent by extending the lines CBSE 2024, 1 mark

SOLUTION

The two lines are not parallel, so when extended they meet at exactly one point.

A pair of lines that intersect at one point has a unique solution and is consistent.

Answer: (a)

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